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Turbine engine with centrifugal compressor having impeller backplate offtake

Abstract

A gas turbine engine includes a fan, a compressor, a combustor, and a turbine. The compressor compresses gases entering the gas turbine engine. The combustor receives the compressed gases from the compressor and mixes fuel with the compressed gases. The turbine receives the hot, high pressure combustion products created by the combustor by igniting the fuel mixed with the compressed gases. The turbine extracts mechanical work from the hot, high pressure combustion products to drive the fan and compressor.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of and claims priority to and the benefit of U.S. patent application Ser. No. 17/747,960, filed 18 May 2022, the disclosure of which is now expressly incorporated herein by reference, which is a continuation of and claims priority to and the benefit of U.S. patent application Ser. No. 16/824,171, filed 19 Mar. 2020, patented as U.S. Pat. No. 11,525,393, issued 13 Dec. 2022, the disclosure of which is now expressly incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to gas turbine engines and more specifically to gas turbine engines with centrifugal compression.

BACKGROUND

(2) Gas turbine engines are used to power aircraft, watercraft, power generators, and the like. Gas turbine engines typically include a compressor, a combustor, and a turbine. The compressor compresses air drawn into the engine and delivers high pressure air to the combustor. In the combustor, fuel is mixed with the high pressure air and is ignited. Products of the combustion reaction in the combustor are directed into the turbine where work is extracted to drive the compressor and, sometimes, an output shaft. Left-over products of the combustion are exhausted out of the turbine and may provide thrust in some applications.

(3) Some gas turbine engines include compressors with one or more stages of axial and/or centrifugal compression. Cooling or avoiding heat generation in compressors may improve component lifetime and performance. Some compressors may be cooled using bleed air that is subsequently discarded, sacrificing both the temperature and pressure of that cooling air.

SUMMARY

(4) The present disclosure may comprise one or more of the following features and combinations thereof.

(5) A turbine engine may include a centrifugal compressor, a combustor, a turbine, and a manifold. The combustor may be fluidly coupled to the centrifugal compressor to receive the majority of compressed gases discharged from the compressor. The turbine may be fluidly coupled to the combustor to receive the hot, high pressure combustion products from the combustor. The turbine may be configured to extract mechanical work from the hot, high pressure combustion products. The manifold may be mounted to the compressor and shaped to define a number of circumferentially spaced apart channels.

(6) In some embodiments, the centrifugal compressor may include an impeller and an impeller backing plate. The impeller may be mounted for rotation about an axis and formed to have impeller blades and an impeller disk. The impeller blades may extend from a forward side of the impeller disk. The impeller backing plate may be arranged along an aft side of the impeller disk to define a back cavity of the centrifugal compressor therebetween.

(7) In some embodiments, the impeller backing plate may be formed to include a number of bleed holes. The bleed holes may be arranged radially inwardly from an outlet tip of the impeller blades so as to allow for compressed gases discharged from the outlet tip of the impeller blades to move over the outlet tip and radially along a portion of the impeller disk on the aft side of the impeller disk before moving through the impeller backing plate,

(8) In some embodiments, the combustor may receive the majority of compressed gases discharged from the outlet tip of the impeller blades as compressor discharge air. The combustor may be configured to mix fuel with the compressor discharge air and ignite the fuel so as to create hot, high pressure combustion products.

(9) In some embodiments, the turbine includes a rotor, at least one set of turbine blades, and a seal. The rotor may have a shaft mounted for rotation about the axis. The set of turbine blades may be coupled to the shaft for rotation therewith. The seal may be formed from forward and aft sealing elements. The forward and aft sealing elements may engage the impeller backing plate of the centrifugal compressor to fluidly separate the back cavity of the compressor from the turbine.

(10) In some embodiments, the forward and aft sealing elements of the seal may form a seal cavity there between. The seal cavity may be in fluid communication with the combustor to receive compressor discharge air from the combustor. In some embodiments, the seal cavity may be pressurized by the compressor discharge air.

(11) In some embodiments, the manifold may be mounted to the impeller backing plate to form the number of circumferentially spaced apart channels. The number of circumferentially spaced apart channels that receive compressed gases moving through the number of bleed holes may carry the compressed gases from the centrifugal compressor to the turbine bypassing the combustor.

(12) In some embodiments, the forward sealing element of the seal included in the turbine may be configured to leak compressor discharge air to the back cavity formed between the impeller disk of

the compressor and the impeller backing plate. The aft sealing element of the seal included in the turbine may be configured to leak compressor discharge air to turbine. In some embodiments, the forward and aft sealing elements may be positioned at the same radial location relative to the shaft of the turbine.

(13) In some embodiments, the impeller backing plate may be formed to include a plurality of impingement holes. The impingement holes may extend radially through the impeller backing plate and open into the seal cavity to transmit compressor discharge air from the compressor to the seal cavity. In some embodiments, the plurality of impingement holes may be circumferentially located between the number of circumferentially spaced apart channels formed by the manifold.

(14) In some embodiments, at least one bleed hole of the number of bleed holes may be configured to provide compressed gases to at least one channel of the number of circumferentially spaced apart channels formed by the manifold. In some embodiments, at least two bleeds holes may be configured to provide compressed gases to at least one channel of the number of circumferentially spaced apart channels formed by the manifold.

(15) In some embodiments, the number of bleed holes may be positioned at a preselected radial location relative to the axis. The preselected radial location of the bleed holes may be positioned so as to bleed an amount of compressed gases from the outlet tip of the impeller blades that decreases windage at or near the outlet tip of the impeller blades.

(16) According to another aspect of the present disclosure, the gas turbine engine may include a centrifugal compressor, a turbine, and a manifold. The centrifugal compressor may include an impeller mounted for rotation about an axis and an impeller backing plate. The impeller may have impeller blades that extend from a forward side of an impeller disk. The impeller backing plate may be arranged along an aft side of the impeller disk to define a back cavity of the centrifugal compressor therebetween.

(17) In some embodiments, the impeller backing plate may be formed to include a number of bleed holes. The number of bleed holes may be arranged radially inwardly from an outlet tip of the impeller blades.

(18) In some embodiments, the turbine may include a rotor and a seal. The rotor may have a shaft mounted for rotation about the axis and a seal. The seal may be formed from forward and aft sealing elements that engage the impeller backing plate of the centrifugal compressor to fluidly separate the back cavity of the compressor from the turbine.

(19) In some embodiments, the manifold may be mounted to the impeller backing plate. The manifold may be shaped to define a number of circumferentially spaced apart channels. The channels may receive compressed bleed gases moving through the number of bleed holes and carry the compressed bleed gases from the centrifugal compressor to the turbine bypassing other components of the gas turbine engine.

(20) In some embodiments, the gas turbine engine may further include a combustor. The combustor may be fluidly coupled to the centrifugal compressor to receive the majority of compressed gases from the centrifugal compressor. The combustor may also be fluidly coupled to the turbine to discharge hot, high pressure combustion products to the turbine.

(21) In some embodiments, the seal cavity may be pressurized. The seal cavity may be pressurized by the compressor discharge air.

(22) In some embodiments, the forward sealing element of the seal may be configured to leak compressor discharge air to the back cavity formed between the impeller disk of the compressor and the impeller backing plate. The aft sealing element of the seal included in the turbine may be configured to leak compressor discharge air to the seal cavity. In some embodiments, the forward and aft sealing elements may be positioned at the same radial location relative to the shaft of the turbine.

(23) In some embodiments, the impeller backing plate may be formed to include a plurality of impingement holes. The plurality of impingement holes may extend radially through the impeller

backing plate and open into the seal cavity to transmit compressed gases moving through the number of circumferentially spaced apart channels. In some embodiments, the plurality of impingement holes may be circumferentially spaced apart between the number of circumferentially spaced apart channels formed by the manifold.

(24) In some embodiments, at least one bleed hole of the number of bleed holes may be configured to provide compressed gases to at least one channel of the number of circumferentially spaced apart channels formed by the manifold. In some embodiments, at least two bleed holes may be configured to provide compressed gases to at least one channel of the number of circumferentially spaced apart channels formed by the manifold.

(25) These and other features of the present disclosure will become more apparent from the following description of the illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cut away perspective view of a gas turbine engine showing the engine includes a fan, an axi-centrifugal compressor, a combustor fluidly coupled to the compressor, and a turbine fluidly coupled to the combustor;

(2) FIG. 2 is a cross section view of the gas turbine engine of FIG. 1 showing the axi-centrifugal compressor has a centrifugal compressor that includes (i) an impeller mounted for rotation about an axis having impeller blades that extend from a forward side of an impeller disk and (ii) an impeller backing plate arranged along an aft side of the impeller disk to define a back cavity of the centrifugal compressor, and further showing the gas turbine engine also includes a manifold coupled to the impeller backing plate that defines a number of channels that transmit a portion of the compressed gases from the centrifugal compressor stage of the compressor to the turbine bypassing the combustor;

(3) FIG. 3 is a detail view of FIG. 2 showing the impeller backing plate is formed to include a number of bleed holes that allow for compressor discharge air from an outlet tip of the impeller blades to move over the outlet tip and radially along a portion of the impeller disk before moving through the number of channels along the impeller backing plate; and

(4) FIG. 4 is a cross section view of the gas turbine engine of FIGS. 1-3 at a different circumferential location showing the turbine includes (i) a rotor having a shaft mounted for rotation about the axis, (ii) at least one set of turbine blades coupled to the shaft for rotation therewith, and (iii) forward and aft sealing elements coupled to the rotor that engage the impeller backing plate to fluidly separate the back cavity of the compressor from the turbine, and further showing impeller backing plate further includes a number of impingement holes that open into a seal cavity defined by the forward and aft sealing elements to transmit compressed gases into the seal cavity.

(5) FIG. 5 is a detail view of one bleed hole included in the number of bleed holes of FIG. 3 showing each of the bleed holes has a conical shape that extends through the impeller backing plate so that an inlet opening opens on a forward surface of the impeller backing plate and an outlet opening opens on an aft surface of the impeller backing plate; and

(6) FIG. 6 is a cross section view of FIG. 5 taken along line 6-6 showing the outlet opening of the bleed hole is radially and circumferentially offset from the inlet opening of the bleed hole relative to the axis of the engine.

DETAILED DESCRIPTION OF THE DRAWINGS

(7) For the purposes of promoting an understanding of the principles of the disclosure, reference will now be made to a number of illustrative embodiments illustrated in the drawings and specific language will be used to describe the same.

(8) An illustrative gas turbine engine 10 includes a fan 12, an axi-centrifugal compressor 14, a

combustor **16** fluidly coupled to the compressor **14**, and a turbine **18** fluidly coupled to the combustor **16** as shown in FIGS. **1-4**. The fan **12** is driven by the turbine **18** and provides thrust for propelling an aircraft. The compressor **14** compresses gases entering the engine **10** along a core flow path (represented by arrows **19**). The compressor **14** delivers the compressed gases to the combustor **16**. The combustor **16** mixes fuel with the compressed gases and ignites the fuel to produce hot, high pressure combustion products. The hot, high pressure combustion products of the combustion reaction in the combustor **16** are directed into the turbine **18** to cause the turbine **18** to rotate about an axis **11** of the gas turbine engine **10**. The turbine **18** extracts mechanical work from the hot, high pressure combustion products to drive the compressor **14** and the fan **12**.

(9) The axi-centrifugal compressor **14** has axial compression stages **13** and a centrifugal compression stage **15** as shown in FIG. **1**. The centrifugal compression stage **15** includes an impeller **22**, an impeller shroud **24**, and an impeller backing plate **26** as shown in FIGS. **2-4**. The impeller **22** is mounted for rotation about the axis **11** and formed to have impeller blades **30**. The impeller blades **30** each extend from a forward side **34** of an impeller disk **32**. The impeller shroud **24** restricts gases from moving over the impeller blades **30** without interacting with the impeller blades **30**. The impeller backing plate **26**, or sometimes referred to herein as an inner combustor case, is arranged along an aft side **36** of the impeller disk **32** to define a back cavity **38** of the centrifugal compressor **15**. In the illustrative embodiment, the compressor **15** further includes a diffuser **28** coupled to impeller **22** to receive the compressed gases from the impeller **22** and discharge the compressed gases to the combustor **16**.

(10) In some embodiments, compressor impellers can generate high temperatures in operation. High temperatures or heat generation by the compressor impeller may be a result of increased windage at or near the compressor impeller tip. Reducing or avoiding heat generation by compressor impellers in operation may improve impeller function, efficiency, and life.

(11) To reduce the heat generation by the compressor impeller, air may be bled from the compressor impeller, specifically at the impeller tip. The bleed air may minimize the windage generated near the compressor impeller tip on the aft side of the impeller disk, yet, due to pressure losses, the bleed air from the compressor impeller may not be useful. Therefore, the bleed air may be subsequently discarded, sacrificing both the temperature and pressure of that bleed air.

(12) As such, the present disclosure teaches an arrangement that reduces the heat generation by the compressor impeller **22**, while allowing for the bleed air to be reused in other components of the gas turbine engine **10**. By reusing the bleed air, the operational efficiency of the gas turbine engine **10** may be increased while maintaining improved impeller operation and life.

(13) To reuse the bleed air from the compressor impeller **22**, the gas turbine engine **10** further includes a manifold **20** as shown in FIGS. **2** and **3**. The manifold **20** is mounted to the impeller backing plate **26** of the compressor **14** to define a number of circumferentially spaced apart channels **42**. The channels **42** receive a bleed flow of compressed gases (represented by arrows **21**) moving through a number of bleed holes **44** formed in the backing plate **26**.

(14) The bleed holes **44** are arranged radially inwardly from an outlet tip **40** of the impeller blades **30** so as to allow the compressed gases discharged from the outlet tip **40** of the impeller blades **30** to move over the outlet tip **40** and radially along a portion of the impeller disk **32** on the aft side **36** before moving through the impeller backing plate **26** as shown in FIGS. **2** and **3**. By bringing some of the compressed gases radially inward along the aft side **36** of the impeller disk **32**, the windage heat generation due to rotation of the impeller disk **32** near the outlet tip **40** may be reduced.

(15) In the illustrative embodiment, the bleed holes **44** are positioned at a preselected radial location **R1** as shown in FIG. **3**. The preselected radial location **R1** of the bleed holes **44** is configured to bleed an amount of compressed gases from the outlet tip **40** of the impeller blades **30** that decreases windage heat generation near the outlet tip **40** of the impeller blades **30** on the aft side **36** of the disk **32**, but also maintains a high pressure capable of being reused in other components of the engine **10**. The channels **42** then carry the compressed gases from the

centrifugal compressor **14** to the turbine **18** bypassing the combustor **16**.

(16) The arrangement of the bleed holes **44** therefore provides a balance between reducing windage heat generated, while also providing a flow of compressed gases with a high enough pressure so as to be reused in other components of the engine **10**. The channels **42** formed by the manifold **20** transport the high pressure compressed gases to the turbine **18** bled from the compressor so that the high pressure compressed gases may be reused in the turbine **18**.

(17) In other embodiments, the compressed gases bled from the outlet tip **40** may be transported to another component of the engine **10**, such as turbine vanes **50** in the turbine **18**. The compressed gases may be transported outside of the engine casing **51** bypassing the combustor **16** to the turbine **18**. The compressed gases may be used to cool the turbine vanes **50** before being discharged into the core flow path **19**.

(18) In the illustrative embodiment, at least one bleed hole **44** is configured to provide compressed gases to at least one channel **42** formed by the manifold **20**. In other embodiments, at least two bleed holes **44** may be configured to provide compressed gases to at least one channel **42** formed by the manifold **20**.

(19) Turning again to the turbine **18**, the turbine **18** includes a plurality of bladed rotating wheel assemblies **48** and a plurality of static turbine vane rings **50** that are fixed relative to the axis **11** as suggested in FIGS. **1-4**. The hot gases are conducted through the core gas path **19** and interact with the bladed wheel assemblies **48** to cause the bladed wheel assemblies **48** to rotate about the axis **11**. Each turbine vane ring **50** includes a plurality of turbine vanes. The turbine vanes **50** are positioned to direct the gases toward the bladed wheel assemblies **48** with a desired orientation.

(20) Each rotating wheel assembly includes a rotor **52**, at least one set of turbine blades **54**, and a seal **56** as shown in FIGS. **1-4**. The rotor **52** has a shaft **58** mounted for rotation about the axis **11**. The set of turbine blades **54** is coupled to the shaft **56** for rotation therewith. The seal **58** is formed from forward and aft sealing elements **60**, **62** that engage the impeller backing plate **26** of the centrifugal compressor **14** to fluidly separate the back cavity **38** of the compressor **14** from the turbine **18**.

(21) The forward sealing element **60** and the aft sealing element **62** of the turbine **18** form a seal cavity **64** therebetween. The seal cavity **64** is in fluid communication with the combustor **16** to receive compressor discharge air from the compressor **14**. In the illustrative embodiment, the forward and aft sealing elements **60**, **62** are positioned at the same radial location relative to the shaft **58** of the turbine **18**.

(22) The impeller backing plate **26** is also formed to include a plurality of impingement holes **46** as shown in FIGS. **2-4**. The impingement holes **46** extend radially through the impeller backing plate **26** and open into the seal cavity **64** to transmit compressor discharge air from the compressor **14** to the seal cavity **64**.

(23) In the illustrative embodiment, the impingement holes **46** are circumferentially located between the number of circumferentially spaced apart channels **42** formed by the manifold **20**. The compressor discharge air pressurizes the seal cavity **64** so that compressor discharge air leaks across the forward sealing element **60** and the aft sealing element **62**.

(24) In the illustrative embodiment, the forward sealing element **60** of the seal **56** is configured to leak compressor discharge air to the back cavity **38** as shown in FIG. **4**. The air leaked to the back cavity **38** pressurizes the back cavity **38**. The air leaked to the back cavity **38** has a high pressure which allows the gases to flow radially outward.

(25) The aft sealing element **62** of the seal **56** is configured to leak compressor discharge air to a wheel cavity **66** as shown in FIG. **4**. The air leaked across the aft sealing element **62** returns into the core flow path **19** at a location between a first stage vane **50** and a first stage bladed wheel assembly **48** of the turbine **18**. Accordingly, the compressed gases purge and cool the wheel cavity **66** before (re)introduction into the core flow path **19**.

(26) In the illustrative embodiment, the channel **42** opens into the wheel cavity **66** as shown in FIG.

3. The bleed flow of compressed gases **21** is configured to purge the cavity **66** and prevent hot gas ingestions from the flow path **19**. The bleed flow of compressed gases **21** is mixed with the leaked compressor discharge air from the aft sealing element **62** in the cavity **66**.

(27) In the illustrative embodiment, the impeller backing plate **26** includes a radially extending portion **70**, an axially extending portion **72**, and a sealing portion **74** as shown in FIGS. **3** and **4**. The radially extending portion **70** extends radially along the aft side **36** of the impeller disk **32** and is spaced apart from the disk **32** to define a portion of the back cavity **38** therebetween. The axially extending portion **72** extends axially from the radially extending portion and is mounted relative to the turbine **18** to define a portion of the back cavity **38** between the impeller backing plate **26** and the disk **32**. The sealing portion **74** extends radially inward from the axially extending portion **72** to engage the forward and aft sealing elements **60**, **62** of the turbine **18**. In the illustrative embodiment, the forward and aft sealing elements **60**, **62** are knife seals that engage abradable bands **76** coupled to the sealing portion **74** of the backing plate **26** as shown in FIGS. **3** and **4**.

(28) In the illustrative embodiment, the bleed holes **44** extend through the radially extending portion **70** of the impeller backing plate **26** from a forward surface **78** of the impeller backing plate **26** to an aft surface **80** of the impeller backing plate **26** as shown in FIGS. **5** and **6**. Each hole **44** has an inlet opening **82** that opens on the forward surface **78** of the impeller backing plate **26** and an outlet opening **84** that opens on the aft surface **80** of the impeller backing plate **26**.

(29) In the illustrative embodiment, the bleed holes **44** are conical in shape as shown in FIGS. **5** and **6**. The inlet opening **82** is sized to have a first diameter **D1**, while the outlet opening **84** is sized to have a second diameter **D2** that is larger than the first diameter **D1** of the inlet opening **82**. The shape of the holes **44** increases/optimizes offtake static pressure.

(30) In the illustrative embodiment, the openings **82**, **84** have a circular shape in the illustrative embodiment, but may have an ovular shape in some embodiments. In other embodiments, the openings **82**, **84** of the holes **44** may be another suitable shape.

(31) Each of the holes **44** extend axially through the backing plate **26** between the openings **82**, **84** along a hole axis **86** as shown in FIG. **6**. A center **83** of the inlet opening **82** and a center **85** of the outlet opening **84** are located on the hole axis **86**. In the illustrative embodiment, the center **85** of the outlet opening **84** is radially and circumferentially offset from the center **83** of the inlet opening **82** relative to the axis **11** of the engine **10**.

(32) The present disclosure related to centrifugal compressors for use in gas turbine engines **10**. One of the limiting mechanical factor of centrifugal compressors may be the metal temperature of the exducer blade tips **40**. The metal temperature may be dependent on the how the back cavity **38** manages heat generation from windage. In some embodiments, air may be bled off the impeller **22** and allowed to flow radially inward along the aft surface **80** of the backing plate **26** before it is thrown overboard. This reduces windage near the outlet tip **40** along the aft side **36** of the impeller disk **32**, but due to pressure losses, the air is not useful and thus thrown overboard. Conversely, if no air is bled off the outlet tip **40**, the air in the back cavity **38** flows radially outward, increasing or maintaining high windage at or near the outlet tip **40**, but providing high pressure, which could be useful in the turbine **18** for a number of applications.

(33) In the illustrative embodiment, the windage heat generation is due to the rotation of the aft side **36** of the impeller disk **32** near the outlet tip **40**. By arranging the bleed holes **44** radially inward from the outlet tip **40**, the bleed flow moves along the aft side **36** near the outlet tip **40** where the windage heat is generated. In this way, the bleed flow reduces the windage heat generation at the aft side **36** of the impeller disk **32** near or radially inward of the outlet tip **40**.

(34) The present disclosure teaches an impeller **22** with a backing plate **26** with a plurality of bleed holes **44** and a manifold **20** coupled to the backing plate **26** that formed channels **42** to transmit the compressed gases bled off the outlet tip **40** to the turbine **18**. The radial position **R1** of the bleed holes **44** is tuned so that windage heat generated between the aft side **36** of the impeller disk **32** and the impeller backing plate **26** near the outlet tip **40** of the impeller **22** may be reduced while the

compressed gases from the outlet tip **40** maintain a high enough pressure to be useful in the turbine **18**.

(35) The plurality of bleed holes **44** are compound in nature as shown in FIGS. **5** and **6**. The holes **44** have a tangential angle relative to the backing plate **32**. In the illustrative embodiments, the tangential angle may be about 65 degrees. The shape of the holes **44** maximizes the static offtake pressure.

(36) While the disclosure has been illustrated and described in detail in the foregoing drawings and description, the same is to be considered as exemplary and not restrictive in character, it being understood that only illustrative embodiments thereof have been shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected.

(37) The bleed flow of compressed gases **21** flows through the channels **42** formed by the manifold **20** and the impeller backing plate **26**, as shown in FIGS. **2-4**. The channels **42** separate the bleed flow of compressed gases **21** from the core flow path **19** of the compressor discharge air, as shown in FIGS. **2-4**. The channels **42** open into the wheel cavity **66**, as shown in FIG. **3** and suggested in paragraph, so that the bleed flow of compressed gases **21** bypasses the seal cavity **64**. In the wheel cavity **66**, the bleed flow of compressed gases **21** mixes with the compressor discharge air that leaks through the aft sealing element **62** into the wheel cavity **66**, as shown in FIG. **4** and suggested in paragraph.

(38) A portion of the compressor discharge air is transmitted from a plenum surrounding the combustor **16** to the seal cavity **64** through the plurality of impingement holes **46** formed in the impeller backing plate **26**, as shown in FIG. **4**. The portion of the compressor discharge air in the seal cavity **64** may leak through the forward sealing element **60** into the back cavity **38**, as shown in FIG. **4**, and may leak through the aft sealing element **62** into the wheel cavity **66**, as shown in FIG. **4**. The compressor discharge air that flows into the wheel cavity **66** to mix with the bleed flow of compressed gases **21** flows through the seal cavity **64** and the aft sealing element **62** before flowing into the wheel cavity **66**, as shown in FIG. **4**.

(39) As shown in FIG. **4**, the back cavity **38** extends radially along the aft side **36** of the impeller disk **32** and axially along the impeller backing plate **26**. The forward sealing element **60** separates an aft end of the back cavity **38** from the seal cavity **64**, as shown in FIG. **4**. The aft sealing element **62** separates the wheel cavity **66** from the seal cavity **64**, as shown in FIG. **4**.

(40) The impeller backing plate **26** is exposed to the mixture of the leaked compressor discharge air and the bleed flow of compressed gases **21** in the back cavity **38** as shown in FIG. **2**. The impeller backing plate **26** is also exposed to the compressor discharge air in the seal cavity **64** and the mixture of the bleed flow of compressed gases **21** and the leaked compressor discharge air in the wheel cavity **66** as shown in FIG. **2**.

Claims

1. A turbine engine comprising: a centrifugal compressor including (i) an impeller mounted for rotation about an axis and having an impeller disk and impeller blades that extend from a forward side of the impeller disk and (ii) an impeller backing plate arranged along an aft side of the impeller disk to define a back cavity of the centrifugal compressor between the aft side of the impeller disk and the impeller backing plate, the impeller backing plate formed to include a number of bleed holes arranged radially inwardly from an outlet tip of each of the impeller blades to allow a first portion of compressed gases discharged from the outlet tips of the impeller blades to move radially inward along a portion of the aft side of the impeller disk before moving through the number of bleed holes in the impeller backing plate, and wherein a second portion of compressed gases is discharged from the centrifugal compressor as a compressor discharge air into a plenum surrounding a combustor, the combustor fluidly coupled to receive a first portion of the compressor discharge air, a turbine fluidly coupled to the combustor and including a rotor having a shaft

mounted for rotation about the axis, a set of turbine blades coupled to the shaft for rotation therewith, and a seal formed from a first sealing element and a second sealing element, wherein a seal cavity is formed between the first sealing element and the second sealing element of the seal, and a manifold mounted to the impeller backing plate and shaped to define a number of channels, wherein the impeller backing plate is formed to include a plurality of impingement holes that extend through the impeller backing plate, the plurality of impingement holes opening into the plenum surrounding the combustor and opening into the seal cavity to transmit a second portion of the compressor discharge air from the plenum to the seal cavity, wherein the first sealing element of the seal is configured to leak at least some of the second portion of the compressor discharge air to the back cavity as a first leaked air so that the first leaked air pressurizes the back cavity, wherein the first portion of compressed gases flows between the impeller backing plate and the manifold through the number of channels.

2. The turbine engine of claim 1, wherein each of the first sealing element and the second sealing element engage the impeller backing plate of the centrifugal compressor to fluidly separate the back cavity of the centrifugal compressor from directly communicating with a wheel cavity of the turbine.

3. The turbine engine of claim 2, wherein the second sealing element of the seal included in the turbine is configured to leak at least some of the second portion of the compressor discharge air as a second leaked air to the wheel cavity of the turbine.

4. The turbine engine of claim 3, wherein the second leaked air mixes with the first portion of compressed gases discharged from the outlet tips of the impeller blades in the wheel cavity.

5. The turbine engine of claim 1, wherein the first portion of compressed gases discharged from the outlet tips of the impeller blades flows through the number of bleed holes in the impeller backing plate and into a wheel cavity of the turbine so that the first portion of compressed gases bypasses the seal cavity.

6. The turbine engine of claim 1, wherein the first sealing element and the second sealing element are positioned at a same radial location relative to the axis.

7. A turbine engine comprising: a centrifugal compressor including (i) an impeller mounted for rotation about an axis and having an impeller disk and impeller blades that extend from a forward side of the impeller disk and (ii) an impeller backing plate arranged along an aft side of the impeller disk to define a back cavity of the centrifugal compressor between the aft side of the impeller disk and the impeller backing plate, the impeller backing plate formed to include a number of bleed holes arranged radially inwardly from an outlet tip of each of the impeller blades to allow a first portion of compressed gases discharged from the outlet tips of the impeller blades to move radially inward along a portion of the aft side of the impeller disk before moving through the number of bleed holes in the impeller backing plate, and wherein a second portion of compressed gases is discharged from the centrifugal compressor as a compressor discharge air into a plenum surrounding a combustor, the combustor fluidly coupled to receive a first portion of the compressor discharge air, a turbine fluidly coupled to the combustor and including a rotor having a shaft mounted for rotation about the axis, a set of turbine blades coupled to the shaft for rotation therewith, and a seal formed from a first sealing element and a second sealing element, wherein a seal cavity is formed between the first sealing element and the second sealing element of the seal, and a manifold mounted to the impeller backing plate and shaped to define a number of circumferentially spaced apart channels that receive the first portion of compressed gases moving through the number of bleed holes and carry the first portion of compressed gases from the centrifugal compressor to a wheel cavity of the turbine such that the first portion of compressed gases bypasses the combustor, wherein the impeller backing plate is formed to include a plurality of impingement holes that extend through the impeller backing plate, the plurality of impingement holes opening into the plenum surrounding the combustor and opening into the seal cavity to transmit a second portion of the compressor discharge air from the plenum to the seal cavity,

wherein the first sealing element of the seal is configured to leak at least some of the second portion of the compressor discharge air to the back cavity as a first leaked air so that the first leaked air the back cavity.

8. The turbine engine of claim 7, wherein the first portion of compressed gases from the centrifugal compressor flows at a radial location that is outward of the seal cavity, to bypass the seal cavity.

9. The turbine of claim 7, wherein at least some of the second portion of the compressor discharge air leaks through the second sealing element into the wheel cavity of the turbine as a second leaked air, and the first portion of compressed gases and the second leaked air mix in the wheel cavity of the turbine.

10. The turbine of claim 7, wherein the first portion of compressed gases flows between the impeller backing plate and the manifold through the number of circumferentially spaced apart channels.

11. The turbine engine of claim 7, wherein each of the first sealing element and the second sealing element engage the impeller backing plate of the centrifugal compressor to fluidly separate the back cavity of the centrifugal compressor from directly communicating with the wheel cavity of the turbine.

12. The turbine engine of claim 7, wherein the first sealing element and the second sealing element are positioned at a same radial location relative to the axis.
